

ENTERPRISE SPECIFICATION SUITE PRD • TRD • ERD • SYSTEM ARCHITECTURE

GroundUp AI Enterprise Specification

Product Requirements Document (PRD) & Technical Requirements Document (TRD) for Developer-Side Financial Visibility, Automated Draw Processing, and Construction Risk Attribution.

DOCUMENT IDENTIFIER	PRODUCT NAME
PRD-TRD-GROUNDUP-2026-V2.2	GroundUp AI Enterprise Platform
STATUS & MATURITY	PRIMARY CONTROL LAYER
PRODUCTION ARCHITECTURE	Real Estate Developer / Project Sponsor
BENCHMARK ASSETS	REGULATORY COMPLIANCE
73 Broadway, 392 1st St, 161 Woodlawn	Cal. Civ. Code §§ 8132-8138 AIA G702 / G703

STRATEGIC VISION

Executive Summary and Problem Statement

GroundUp AI helps real estate developers know where every dollar is going, where every delay is coming from, and whether project profit is at risk. It is the owner-side control layer for financial visibility, draw tracking, delay attribution, and profit-risk reporting.

The Core Industry Problem

Mid-market real estate developers managing \$5M to \$50M projects face fragmented, disconnected data. Construction budgets, field progress photos, bank loan draw packages, and accounting spreadsheets exist in separate silos. Project sponsors frequently discover cost overruns and loan interest escalation 60 to 90 days after trade contractors incur the expense. This lag erodes project profitability and threatens loan covenants.

The GroundUp AI Solution

GroundUp AI provides a continuous financial operating system. By harmonizing automated Document AI extraction, multi-source bank reconciliation, and timestamped field verification, GroundUp maintains Four Synchronized Truths: Budget Truth, Spend Truth, Progress Truth, and Funding Truth. Developers, finance controllers, and capital partners gain real-time visibility into forecast at completion, loan interest burn, and realized investor returns.

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Live ERD Explorer (28 Tables)

Practical, real-world operational benchmarks designed for ground-up construction management.

1. Draw Package Turnaround

Same-Day (vs weeks of manual prep)

Assemble audit-ready AIA G702/G703 draw packages with verified invoices, retainage ledgers, and lien waivers in hours rather than weeks of manual compilation.

2. Document Ingestion Efficiency

Automated (with human review)

Extract line items, vendor details, and amounts across any document format or ZIP archive, reducing manual invoice entry while keeping human review in the loop.

3. Lien & Title Risk Prevention

Proactive (statutory protection)

Track California Civil Code Section 8132 and 8134 conditional and unconditional lien waivers alongside every payment to protect against mechanics liens and title disputes.

STAKEHOLDER CONSENSUS

1. Roles at a Glance and Stakeholder Architecture

Role definitions finalized for the owner-side control layer. The owner decides which general contractor workflow applies to each development project.

ROLE	TYPE	MAIN RESPONSIBILITY	PLATFORM ACCESS LEVEL
Owner / Developer	INTERNAL	Provides all documents, chooses the GC and GC type, approves change orders, tracks missing items, builds the draw packet, shares investor previews.	FULL ACCESS
CFO / Accounting	INTERNAL	Provides the expense sheet, matches expenses to budget lines, reconciles draws, exports variance reports.	FINANCIAL MODULES

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Project Manager	INTERNAL	Updates milestones, logs inspection status, tracks delay sources and carry costs.	TIMELINE & EVIDENCE
Investor / Partner	EXTERNAL	Views project health, next funding event, and monthly update. Read-only summaries without subcontractor-level detail.	READ-ONLY
GC (Fixed Contract)	EXTERNAL	Submits milestone claims, completion proof, and change orders. Paid on milestone completion without exposing subcontractor bills.	LIMITED SUBMISSION
GC (Daily Updates)	EXTERNAL	Posts daily updates, invoices and costs, daily photos, and change orders. Open-book tracking with GC markup displayed separately.	LIMITED SUBMISSION

Governance Rule: Ownership of Administrative Work

The owner decides which GC type applies to each project. Admin or coordinator work (collecting documents, tracking missing items, building the draw packet) belongs to the owner. In v1, the general contractor is an input source rather than the primary customer.

Architectural Principle: Why the Lender Distinction Matters

The construction lender is intentionally designed **not** to be an active interactive platform user initially. Instead, GroundUp receives and ingests lender data.

Commercial banks operate behind strict IT firewalls and compliance protocols. Forcing a loan officer to register a user account, remember another password, and log into a third-party software creates friction and delays draw approvals. GroundUp treats the lender as an external inbound data source (ingesting loan approvals, inspection waivers, and Columbia Bank draw statements) and an outbound delivery target (producing fully audit-ready AIA G702/G703 draw packages). This guarantees immediate adoption without waiting for bank IT approvals.

FUNCTIONAL SCOPE

2. Scope Prioritization and User Stories

Functional breakdown categorized by MoSCoW prioritization and engineering priority tiers.

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US-01	Owner / Developer	As a Developer, I want to view my live project profit and forecast at completion, so I can detect cost overruns before they erode my equity.	Must Have	P1	Executive dashboard displays baseline pro forma, actual spend, remaining budget, daily interest carry, and dynamic forecast profit.
US-02	CFO / Accounting	As a CFO, I want invoices, receipts, and draw packets in any format automatically ingested and matched to budget lines, so I can eliminate manual data entry.	Must Have	P1	Document AI automatically unpacks ZIP archives and extracts vendor, date, line items, gross amount, and retainage with split-screen human review.
US-03	Owner / Developer	As an Owner, I want to assemble draw applications with attached lien waivers, so our contractors get paid without administrative delays.	Must Have	P1	System bundles approved invoices, checks California Civil Code Section 8132 waivers, and generates AIA G702/G703 PDF packets.
US-04	Project Manager	As a Project Manager, I want to upload photos and municipal inspection cards from my phone, so field work is immediately verified for draw release.	Must Have	P1	Mobile app uploads geotagged, timestamped site photos linked directly to specific milestone cost codes.
US-05	Investor / Partner	As an Investor, I want to view my invested capital balance and expected return, so I have transparent oversight without emailing the sponsor.	Should Have	P2	Investor portal displays equity deployed, project progress percentage, quarterly distribution forecasts, and photo proof.
US-06	CFO / Accounting	As a CFO, I want corporate credit card charges and bank transfers matched to	Should Have	P2	Automated bank reconciliation engine matches bank wires and corporate card

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		invoices, so every penny is reconciled.	charges to verified invoice line items.		
US-07	Owner / Developer	As an Owner, I want to simulate schedule delay impacts on loan interest, so I can make informed trade acceleration decisions.	Could Have	P3	Interactive slider computes daily loan interest burn and revised investor IRR based on days of construction delay.

INDUSTRY COMPLIANCE

3. U.S. Construction Market Assumptions

Configurable financial and statutory parameters reflecting real-world American commercial and residential construction standards.

PARAMETER	STANDARD BASELINE	STATUTORY / INDUSTRY SOURCE	PLATFORM IMPLEMENTATION LOGIC
Retainage Withholding	10.0% standard (5.0% post-50% completion)	AIA Document A201 § 9.3.1	Deducted from gross invoice amounts; tracked in a separate retainage liability ledger until final certificate of occupancy.
Payment Applications	AIA G702 / G703 Master Format	American Institute of Architects	Standardized application and certificate for payment with continuation sheets mapped to CSI MasterFormat divisions.
Progress Lien Waiver (Conditional)	California Civil Code § 8132	State Statutory Standard	Mandatory attachment for every invoice submitted in an active draw package; releases lien rights conditioned on wire clearance.
Progress Lien Waiver (Unconditional)	California Civil Code § 8134	State Statutory Standard	Required from trade contractors upon cleared bank disbursement from the prior billing cycle.
Final Lien Waiver (Conditional)	California Civil Code § 8136	State Statutory Standard	Required with final retainage billing application prior to final loan release.

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Final Lien Waiver (Unconditional)	California Civil Code § 8138	State Statutory Standard	Executed upon final retainage disbursement and project sign-off.
Cost Code Hierarchy	CSI MasterFormat (50 Divisions)	Construction Specifications Institute	Structured cost codes linking budget line items to trade contracts, invoice items, and draw requests.
Daily Debt Carry	Accrued daily on drawn loan balance	Commercial Construction Loan Agreements	Calculated as: Drawn Debt Principal multiplied by (Annual Rate / 365) multiplied by Delay Days.

USE CASES & ROLE ARCHITECTURE

4. Combined Use Case Diagrams and Role Explanations

GroundUp AI acts as the owner-side control layer. The platform keeps four data streams separate and reconciles them in reporting, avoiding the trap of mixing disparate data on one screen.

Core Concept: Four Separate Data Streams Reconciled in Reporting

The platform keeps four data streams separate and reconciles them, instead of mixing them in one screen:

- **Budget is the plan:** Budget sheet and schedule of values lines locked at project baseline.
- **Expenses are actual spend:** Incurred actuals supplied by the CFO as the expense sheet.
- **Draws are lender funding:** Requested, approved, and funded amounts from the construction loan.
- **Timeline compares planned and actual milestones:** Identifies schedule slippage and explains delay attribution.

Combined Architectural Use Case Diagrams

Each project has one general contractor, and the owner chooses the GC type per project. The platform features two combined diagrams: one for a project with a **Fixed Contract GC** and one for a project with a **Daily-Updates GC**. Both diagrams show every active stakeholder: Owner, Project Manager, Investor, and CFO. **Purple marks the internal team** and **teal marks external parties**.

3.1 Project with a Fixed Contract GC

The GC submits milestone claims, completion proof, and change orders. The owner pays on milestones and does not see subcontractor-level invoices.

MILESTONE MODEL

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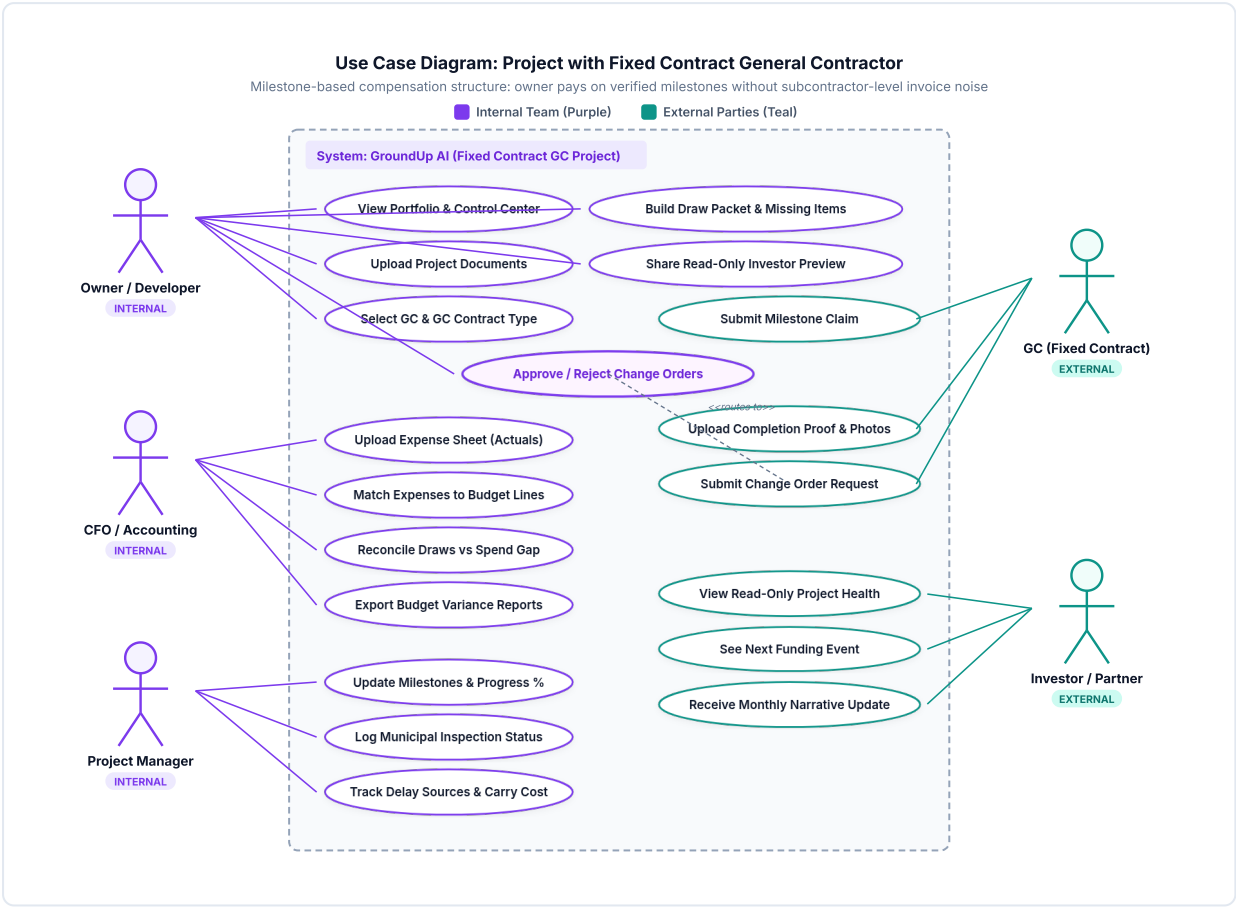


Figure 1: Behavioral interaction model for a fixed or milestone-based GC contract. Purple marks the internal team and teal marks external parties. The owner pays on milestone proof without subcontractor-level invoice noise.

3.2 Project with a Daily-Updates GC

The GC posts daily updates, invoices and costs, and daily photos, plus change orders. The owner sees granular costs with the GC markup shown separately.

COST-PLUS /
OPEN-BOOK

FIGURE 1B Use Case Model: Daily-Updates General Contractor Project

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Live ERD Explorer (28 Tables)

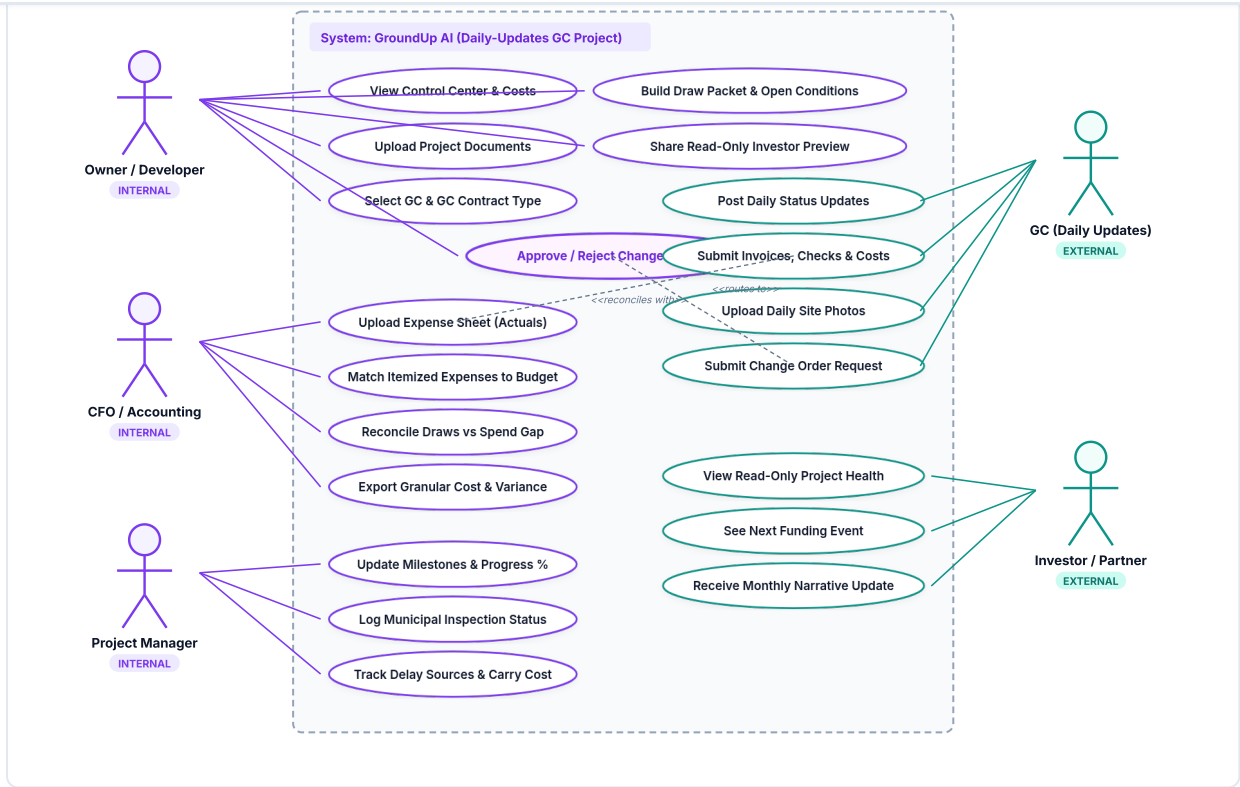


Figure 1b: Behavioral interaction model for cost-plus or open-book projects. Purple marks the internal team and teal marks external parties. GC posts daily progress, invoices, checks, and card costs with GC markup separated.

Detailed Functional Use Cases by Role

Exhaustive functional breakdown defining specific user actions, system outcomes, and data deliverables across all six role categories.

4.1 Owner / Developer (Primary Buyer & Central Role) INTERNAL | FULL ACCESS

Primary Goal: Know financial health, project profit, schedule risk, and what decision to make next. The owner is the primary customer and central role in the platform.

USE CASE	WHAT THE OWNER DOES	OUTCOME
View portfolio dashboard	Opens the home screen with project cards showing status, cash gap, delay days, next draw, and top alerts.	Sees which project needs attention today.
Monitor project control center	Reviews timeline (planned, actual, and expected start and completion), budgeted amount, actual spend, lender funded	One screen shows whether the project is healthy and why it drifts.

	evidence, inspection status, fund status, and units sold.	
Upload project documents	Uploads the foundational project documents listed below.	Source of truth for budget, timeline, and loan data.
Select GC and GC type	Chooses the GC and marks the contract as fixed or daily updates.	Enables the matching GC workflow for that specific development.
Approve change orders	Reviews cost, reason, and impact on budget and timeline, then approves or rejects.	No hidden or inflated change orders enter project liabilities.
Track missing items	Reviews outstanding documents, open conditions, and missing lien waivers.	Nothing blocks a lender draw unnoticed.
Build draw packet	Selects the draw period; system lists eligible budget lines, invoices, photos, and open lender conditions and produces a portal-entry summary.	Clean packet the owner submits through the lender portal.
Share investor preview	Generates the read-only project health view.	Investor confidence without exposing subcontractor-level invoices.

Documents the Owner Provides:

- Budget sheet and Schedule of Values (SOV)
- General Contractor contract (fixed milestone or cost-plus agreement)
- Master construction timeline / Gantt chart
- Bank statements and corporate account ledgers
- Loan approval statement and commercial credit agreement
- HUD settlement statement
- Land acquisition statement and closing binder

4.2 CFO / AccountingINTERNAL | FINANCIAL MODULES

Primary Goal: Accurate expense categories, ledger reconciliation, and budget variance. The CFO provides the expense sheet and executes the accounting workflows that follow from it.

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Upload expense sheet	Imports actuals (vendor, date, amount, description, payment status); the system normalizes vendors, dates, amounts, and categories.	Clean ledger of actual project spend.
Match expense to budget	Confirms or corrects the system match of each expense to a budget line; low-confidence rows go to a review queue.	Budget vs actual and funded vs spent stay current.
Reconcile draws	Compares actual spend with requested, approved, and funded draws.	Cash gap and spend not yet drawn are visible in real time.
Export variance reports	Produces expense sheet, budget variance report, draw history, and summaries as Excel or PDF.	Ready to share with owner, lender, or audit partners.

Strict Financial Guardrail: AI suggests expense matches but never changes financial data without the CFO explicit sign-off.

4.3 Project Manager

INTERNAL | TIMELINE & EVIDENCE

Primary Goal: Trade progress, municipal inspections, and schedule slippage, with empirical evidence behind each field update.

USE CASE	WHAT THE PM DOES	OUTCOME
Update milestones	Records actual start and completion dates, expected completion, progress %, and photo evidence against the planned timeline.	Timeline reflects ground reality on site.
Log inspection status	Marks each inspection as scheduled, passed, failed, or delayed and attaches signed municipal inspection cards.	Inspection risk is immediately visible to leadership.
Track delay sources	Tags the root cause of a slip (plumbing, electrical, permit, municipal inspector availability, GC coordination); system records	Owner sees who or what causes delay and what it costs.

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Live ERD Explorer (28 Tables)

and carry cost.

Production Field Example: An electrical rough inspection planned for February happens in March. The platform tags the delay as 45 days late and immediately links that slip to the daily interest carry cost the owner sees on the executive dashboard.

4.4 Investor / Partner

EXTERNAL | READ-ONLY

Primary Goal: Capital invested, project status, expected ROI, risk profile, and upcoming funding events. Access is read-only and exists only when the owner explicitly shares it.

USE CASE	WHAT THE INVESTOR SEES	OUTCOME
View project health	Budget vs actual summary, timeline health, risk level, and units sold.	Confidence without detail overload or confusion.
See next funding event	The upcoming bank draw or capital call and its expected timing.	Knows exactly when capital moves and when equity yields disburse.
Receive monthly update	Generated narrative update that the owner approves before sending.	Regular, consistent reporting without administrative overhead.

Privacy Boundary: Individual subcontractor invoices, vendor-level bills, and internal trade disputes remain hidden from external investor views.

4.5 GC (Fixed Contract)

EXTERNAL | LIMITED SUBMISSION

Primary Goal: Get paid on agreed milestones. The owner chooses this GC type when the project is a fixed or milestone contract (for example, a \$2M total contract with site work \$300K, framing \$400K, and mechanical \$350K).

USE CASE	WHAT THE GC DOES	OUTCOME
Submit milestone claim	Requests payment when an agreed milestone is certified complete.	Owner sees milestone status and payment due.

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Upload completion proof	Attaches photos and sign-off documents showing the milestone is done.	Payment is backed by empirical evidence and inspection sign-off.
Submit change order	Enters scope description, cost adjustment, and reason for work outside the contract.	Routed directly to the owner for formal approval or rejection.

4.6 GC (Daily Updates)

EXTERNAL | LIMITED SUBMISSION

Primary Goal: Keep the owner informed every day while costs are tracked openly. The owner chooses this GC type for open-book, cost-plus, or joint partnership projects.

USE CASE	WHAT THE GC DOES	OUTCOME
Post daily status update	Reports progress, site issues, and expected milestone completion each day.	Timeline and delay tracking stay current in real time.
Submit invoices, costs	Uploads trade invoices, checks, card transactions, and line-item costs.	Granular actual cost data for CFO reconciliation.
Upload daily photos	Attaches site photos as continuous progress proof.	Evidence for progress % and inspection sign-offs.
Submit change order	Enters scope, cost, and reason for added work outside the original estimate.	Routed to the owner for formal approval.

Fixed Contract GC vs Daily-Updates GC Comparison

DIMENSION	FIXED CONTRACT GC	DAILY-UPDATES GC
Contract Model	Fixed total with scheduled milestone payments (e.g. \$2M contract).	Open-book / cost-plus: actual costs plus the GC markup or profit share.
Owner Visibility	Milestone status and payment due, without subcontractor invoice noise.	Granular costs, with the GC markup shown transparently separated.

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Reconciliation	Paid on milestone completion, backed by site proof and inspection.	Line-item reconciliation of trade invoices, checks, and card transactions.
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A GC never chooses the type; the owner sets it per project. In v1 the GC is an input source, not the primary customer.

CORE FINANCIAL RECORDS

5. The Four Truths Financial Architecture

How GroundUp AI synchronizes approved budgets, actual incurred expenses, verified physical site progress, and bank loan draws to maintain continuous financial clarity.

FIGURE 2 The Four Truths Financial Architecture

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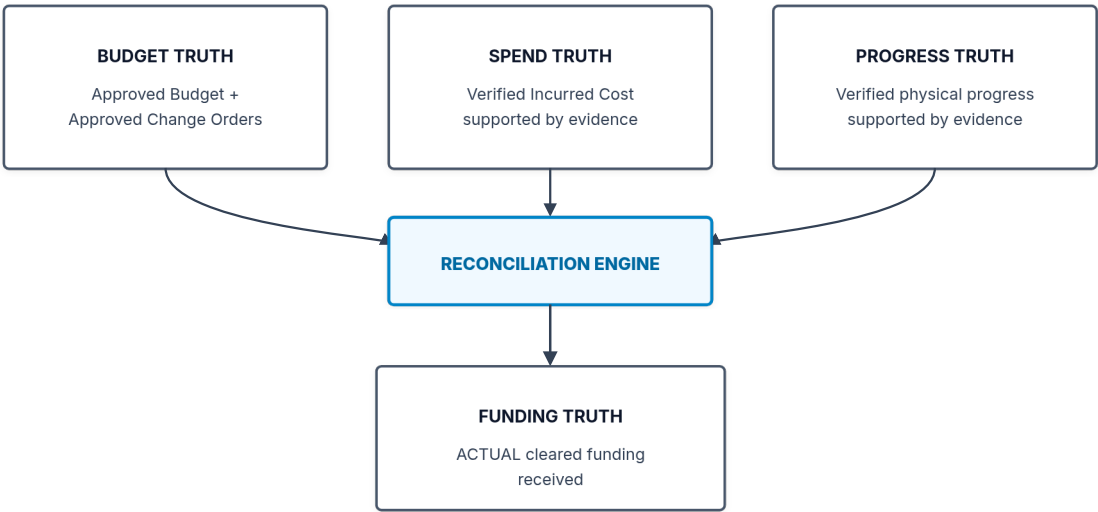


Figure 2: Continuous reconciliation between Budget Truth, Spend Truth, Progress Truth, and Funding Truth.

CASH FLOW MANAGEMENT

6. Capital Flow and Payment Process

Clear flow of funds from investor equity and bank loan draws through 10% retainage holdbacks and payments to trade contractors.

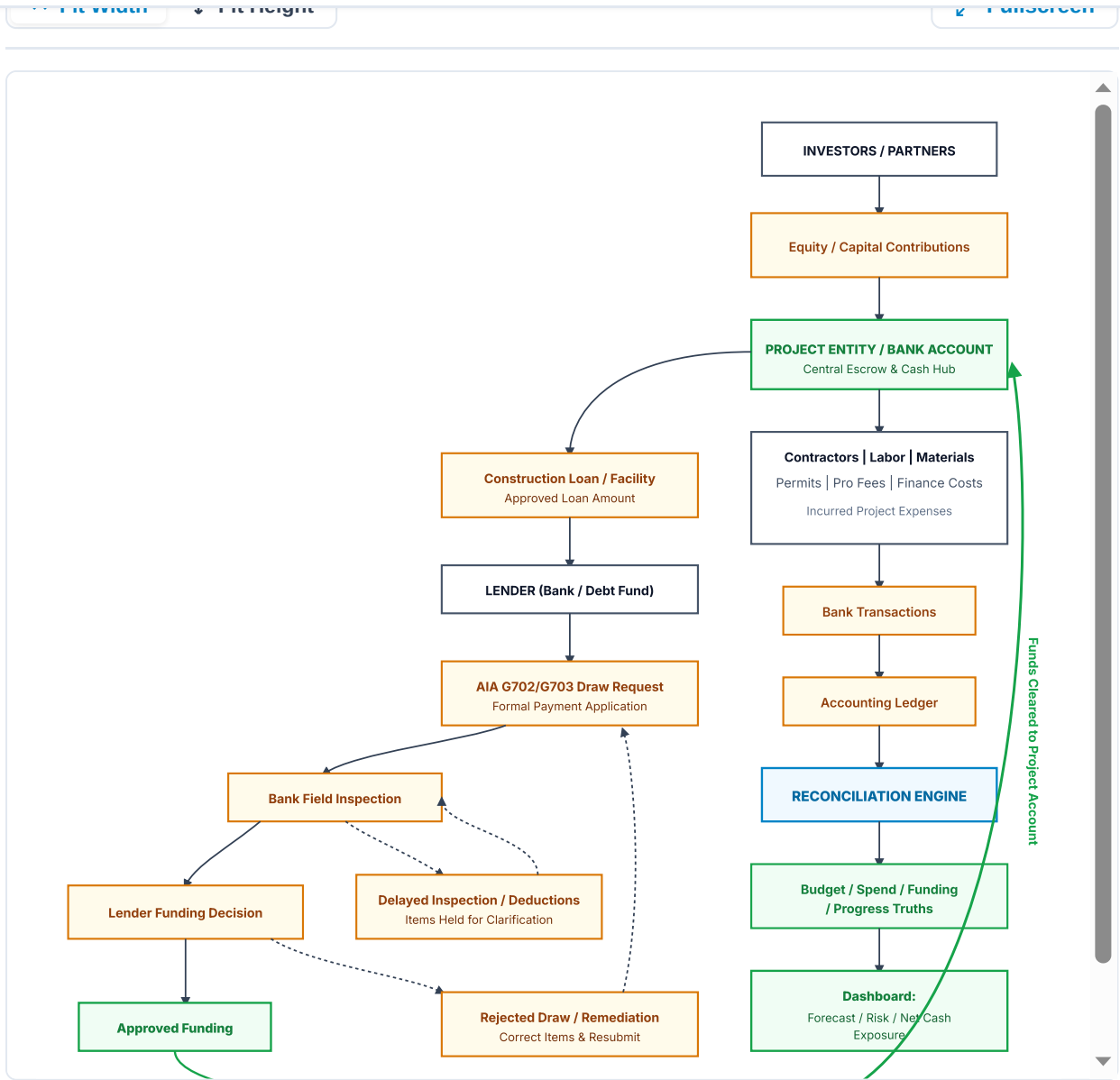


Figure 3: Cash and draw mechanics across developers, lenders, general contractors, and trade subcontractors.

DAILY WORKFLOW

7. Project Financial Lifecycle Workflow

Step-by-step process for document upload, automated data reading, invoice checks, team approvals, and accounting updates.

FIGURE 4 Project Financial Lifecycle Workflow

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GroundUp AI Specification

Live ERD Explorer (28 Tables)

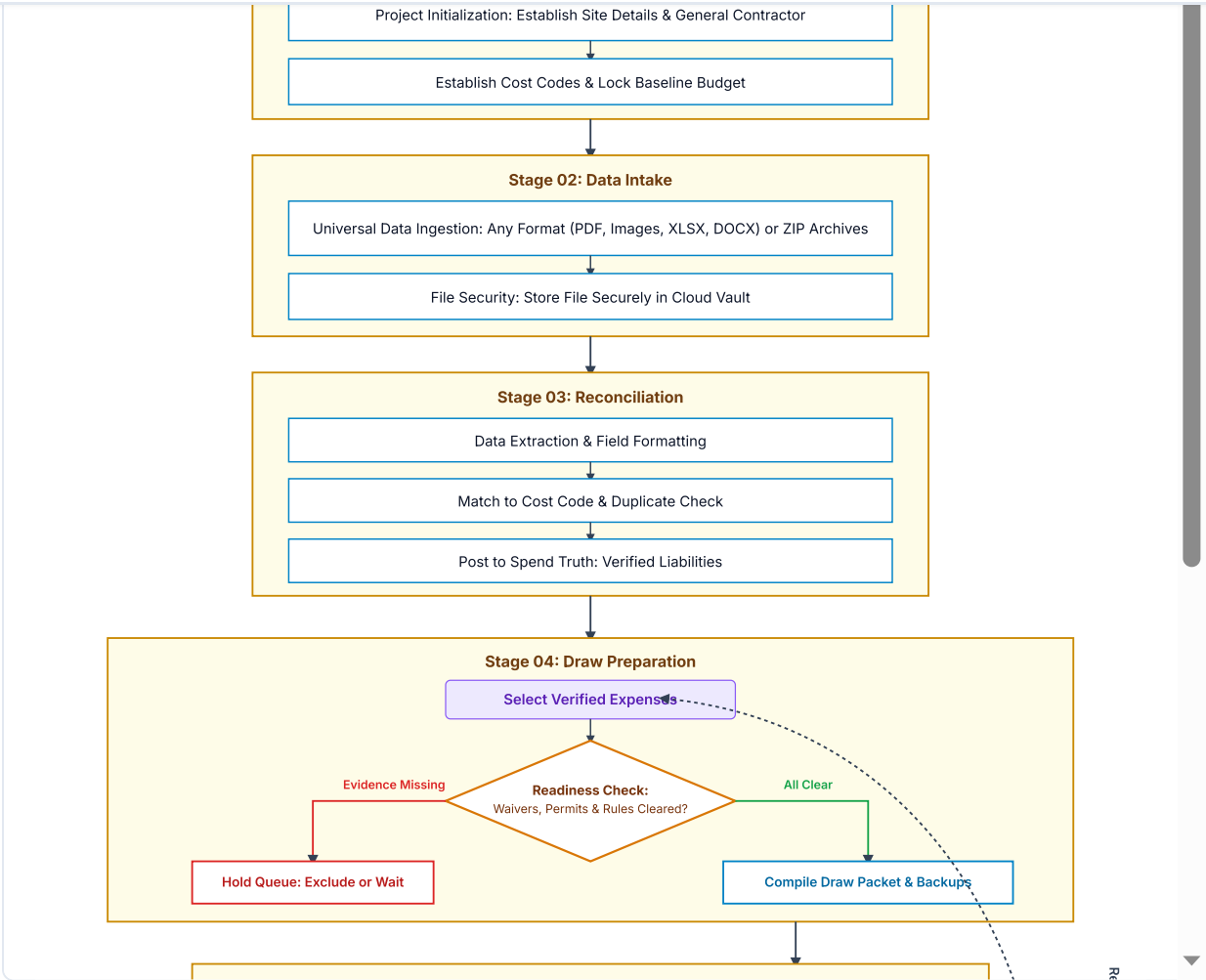


Figure 4: Daily workflow showing document intake, AI extraction, compliance checks, and approval gates.

LOAN DRAW WORKFLOW

8. Bank Loan Draw Process Flow and 6-Step Cycle

Clear stages for preparing bank draw packages, bank inspection sign-offs, handling lender holdbacks, and wire releases.

The 6-Step Draw Cycle Flow

- Documents and spend:** The owner uploads project documents, the CFO uploads the expense sheet, and the GC submits claims or daily updates.
- Progress:** The project manager confirms physical progress and attaches passed inspection sign-offs.
- Matching:** The CFO matches expenses to budget lines and reviews variance.
- Packet:** The owner checks missing items, verifies conditional lien waivers, and builds the draw packet.
- Submission:** The owner submits through the lender portal, and the lender funds the draw wire.

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[Live ERD Explorer \(28 Tables\)](#)

FIGURE 5 Bank Loan Draw Process Flow

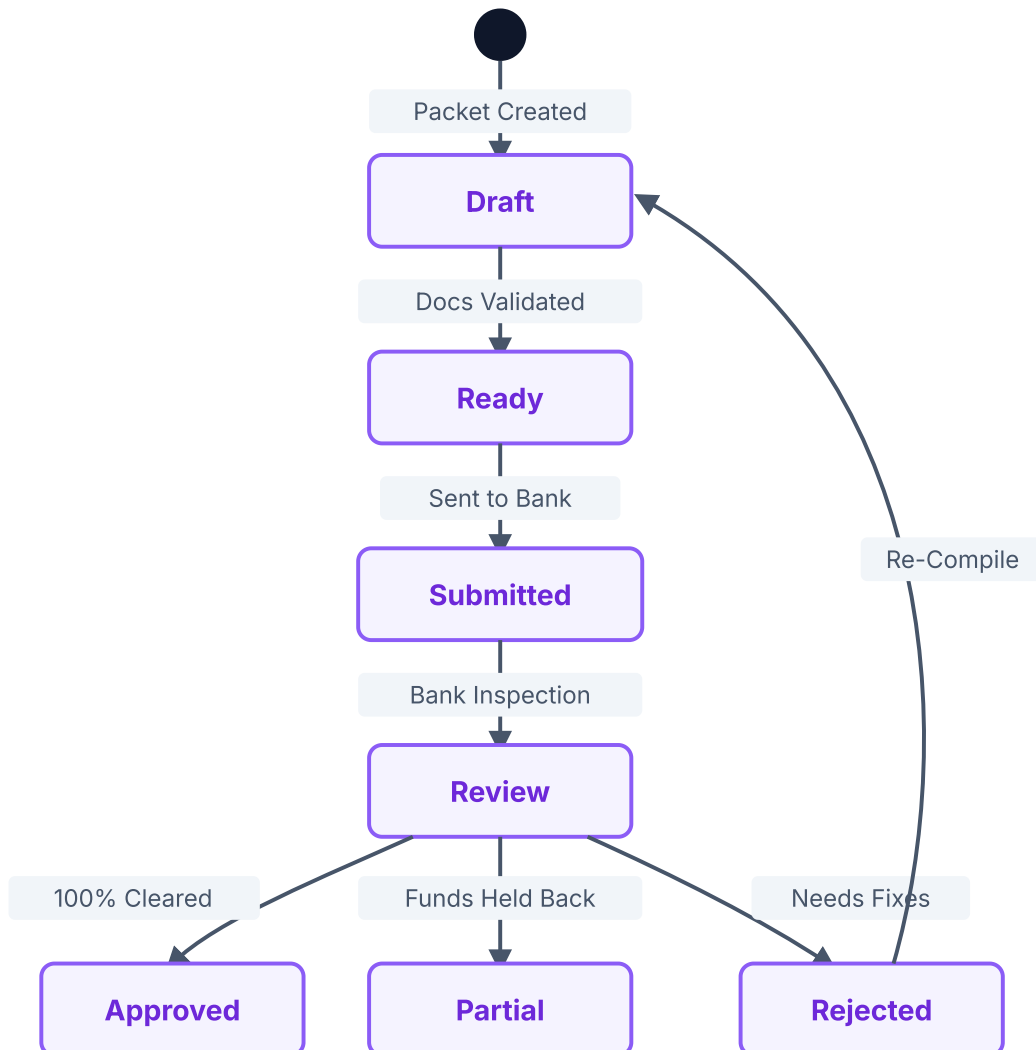
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Figure 5: Step-by-step lifecycle for bank loan draw packages, including revision branching when lenders hold back funds.

Core Platform Business Rules

- **Separate and Reconciled:** Budget, expenses, draws, and timeline stay separate and are reconciled in reporting.
- **Human in the Loop:** AI suggests; humans approve. No financial data changes silently.
- **Lender Integration:** GroundUp AI prepares audit-ready draw packets but does not replace the lender portal.
- **GC Configuration:** The owner selects the GC type per project (Fixed Contract vs Daily Updates).
- **Change Order Authority:** All change orders require formal owner approval before altering budget baselines.

PROJECT SETUP

9. Project Setup and Baseline Budget Flow

How to set up a new project, lock in the initial budget, assign construction cost codes, and enter loan terms.

FIGURE 6 Project Setup and Baseline Budget Flow

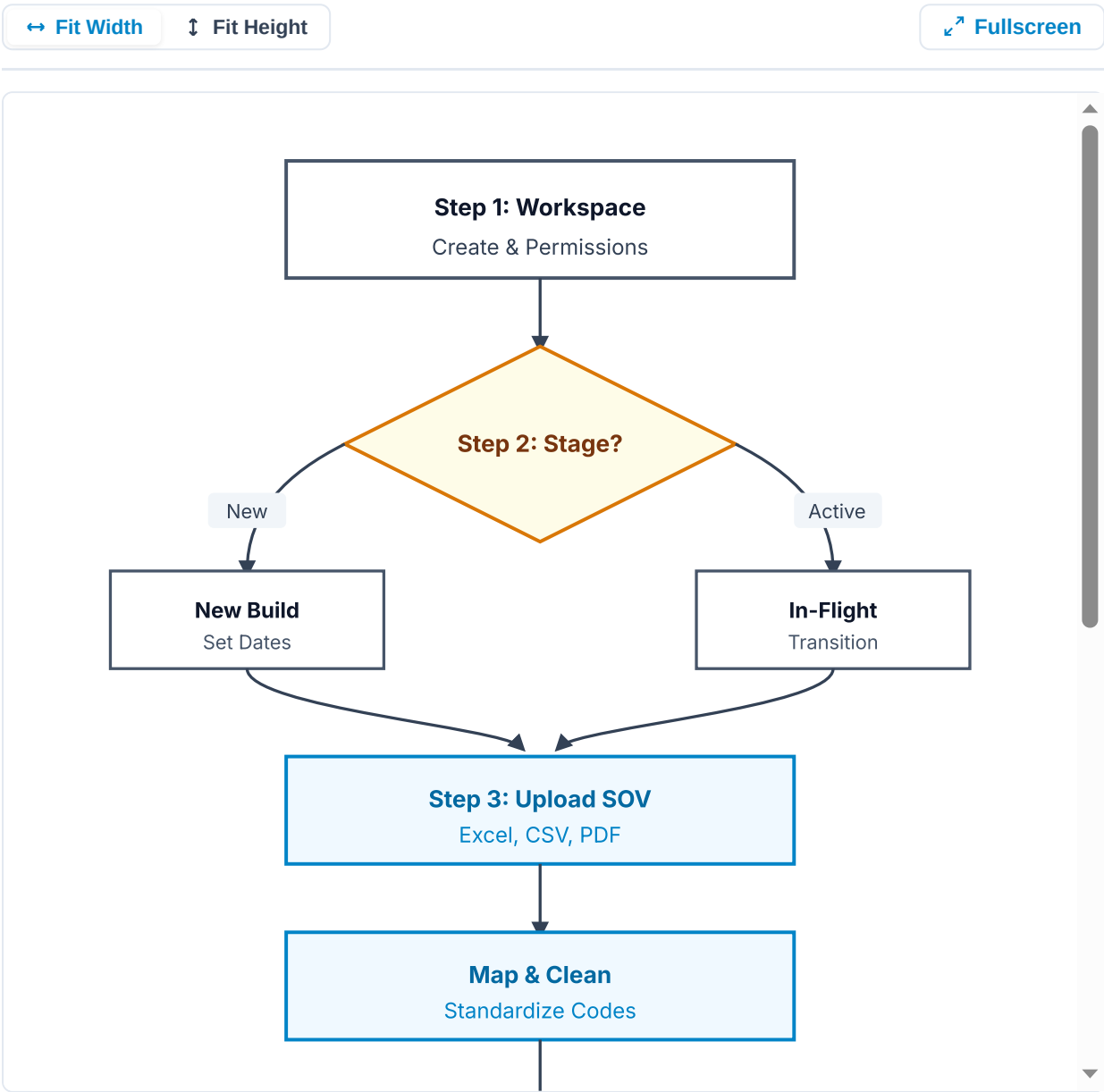


Figure 6: Initialization sequence from project creation to budget lock and baseline financial setup.

QUALITY CONTROL & SAFETY

numbers enter your accounting records.

FIGURE 7 AI Safety Rules and Human Approval Gates

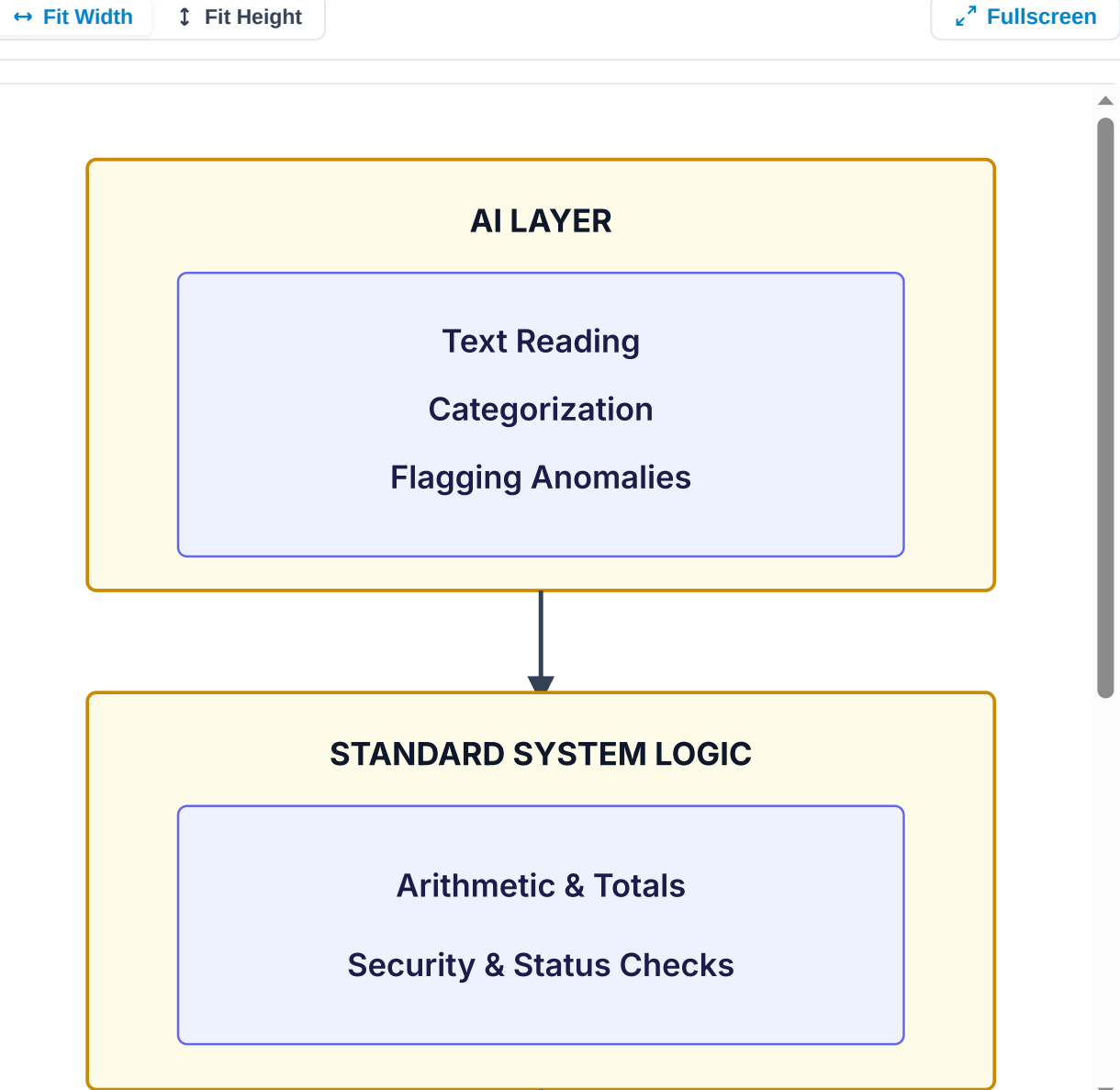


Figure 7: Safety boundaries separating machine data reading from human financial sign-off.

PART II: TECHNICAL REQUIREMENTS DOCUMENT (TRD)

System architecture, multi-project data models, relational master tables, and mathematical forecasting logic.

SYSTEM ARCHITECTURE

11. Nine-Tier Enterprise System Architecture

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Live ERD Explorer (28 Tables)

FIGURE 8 Nine-Tier Enterprise System Architecture

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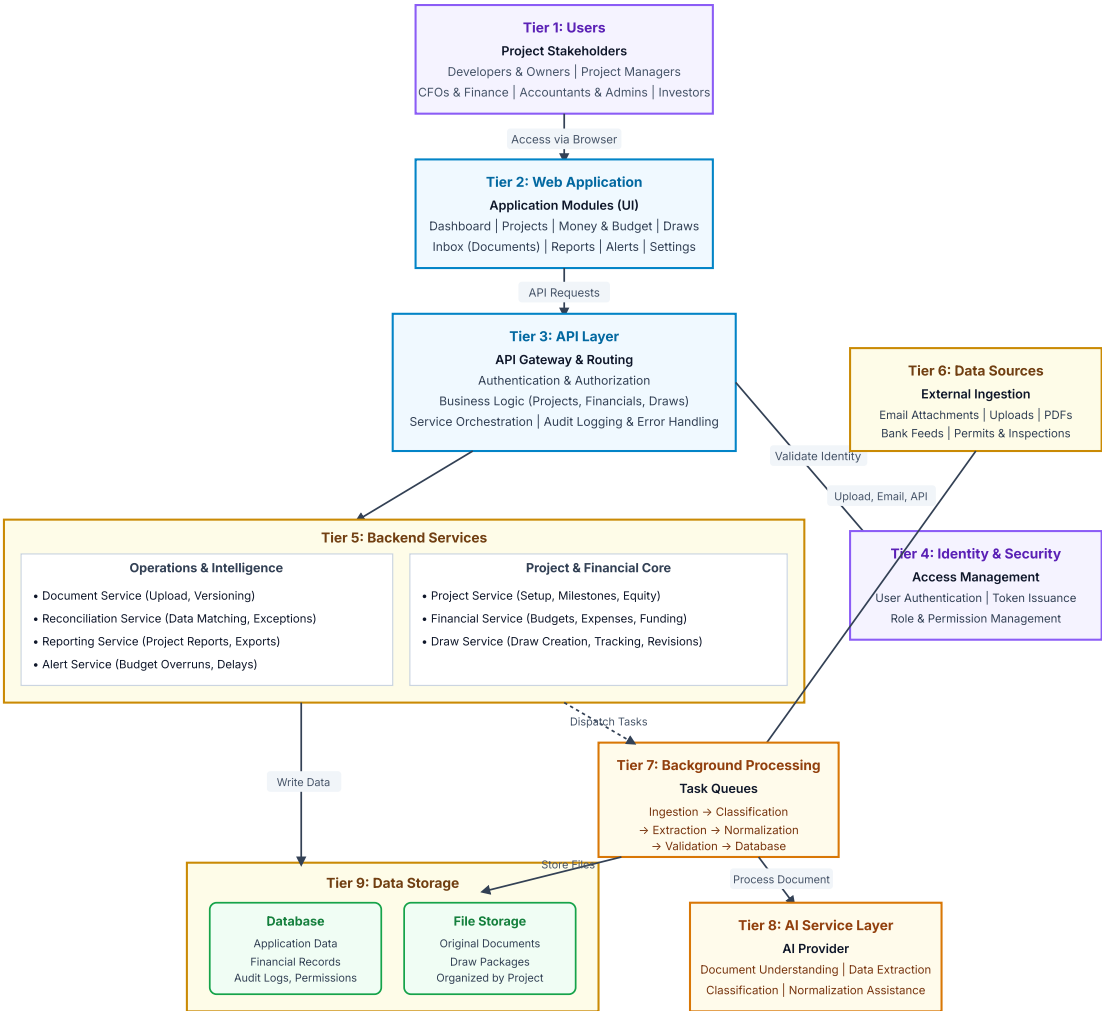


Figure 8: System diagram showing user apps, secure gateway, background workers, and databases.

DOCUMENT AI PIPELINE

12. Ten-Step Document AI Processing Pipeline

Automated optical character recognition, field classification, schema normalization, and confidence-scored human verification supporting all document formats and batch ZIP archives.

Universal Format Ingestion Engine

ANY FORMAT ACCEPTED

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- **Raster & Mobile Images:** JPG, JPEG, PNG, TIFF, HEIC (direct iPhone/Android site photos).
- **Spreadsheets & Data Feeds:** Microsoft Excel (.xlsx, .xls), CSV, TSV (expense ledgers, bank transactions).
- **Word Processing & Text:** Microsoft Word (.docx, .doc), Rich Text (.rtf), Plain Text (.txt).
- **Drawings & Blueprints:** Architectural scans, CAD exports, municipal inspection card scans.

Automated ZIP Archive Unpacking & Batch Processing

BATCH ARCHIVE EXTRACTION

Trade contractors and accountants frequently submit monthly draw documentation as compressed archives:

- **Archive Formats:** .zip, .tar.gz, .7z multi-file packages.
- **Isolated Sandbox Unpack:** Archives are decompressed in an isolated memory buffer with recursive decompression bomb protection.
- **Antivirus & Checksum Audit:** Every extracted file undergoes SHA-256 integrity hashing and malware screening.
- **Folder Hierarchy Preservation:** Subfolder organization (e.g. Invoices/, Lien_Waivers/, Photos/) is preserved as metadata tags.
- **Asynchronous Fan-Out:** Individual constituent documents are scheduled in parallel across the Document AI extraction queue.

FIGURE 9 Ten-Step Document AI Processing Pipeline

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Live ERD Explorer (28 Tables)

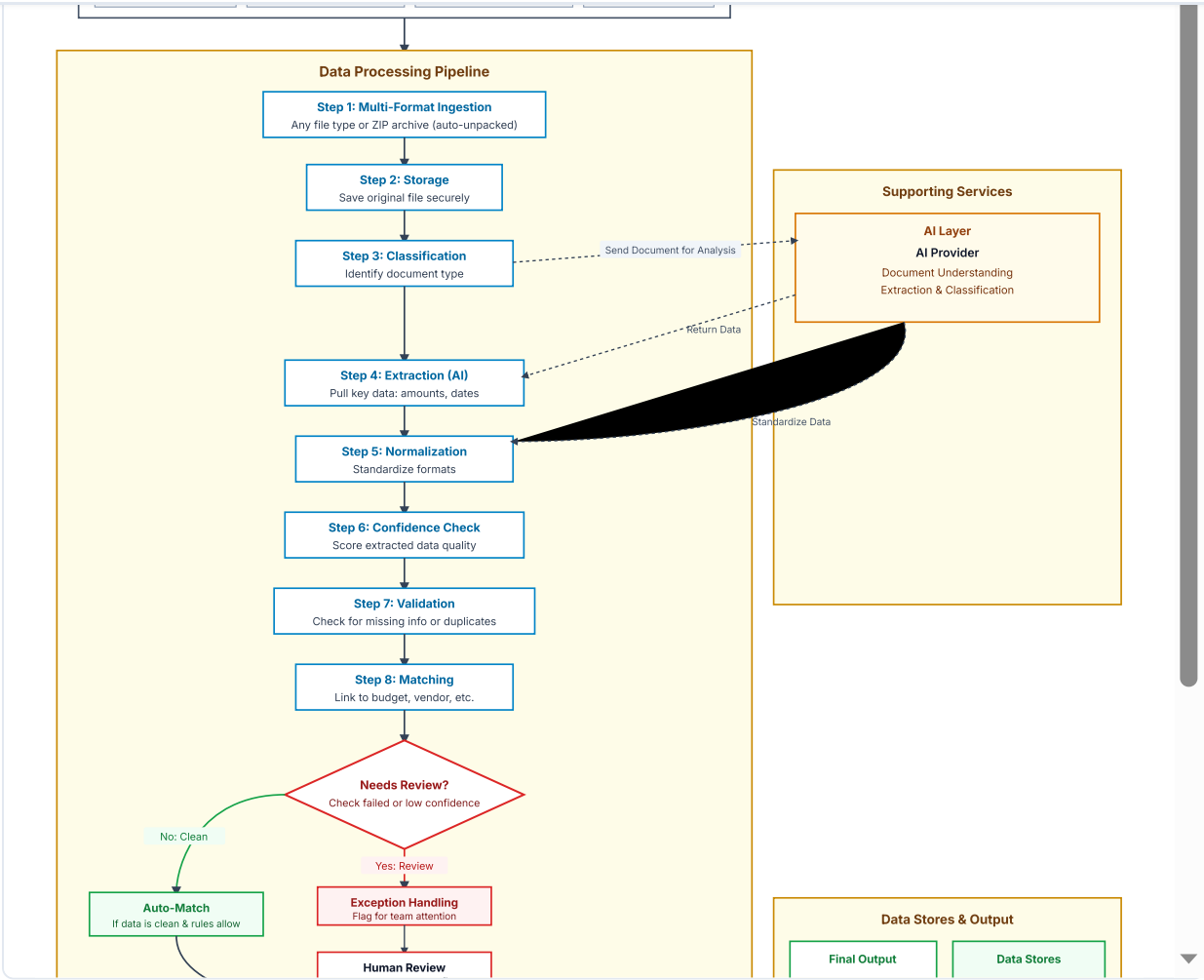


Figure 9: Universal multi-format ingestion pipeline processing any file format or ZIP archive into verified financial records.

DATA MODEL

13. System Data Model

Visual blueprint showing how projects, budgets, trade contracts, invoices, bank draws, and investor returns connect together.

FIGURE 10 System Data Model

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Live ERD Explorer (28 Tables)

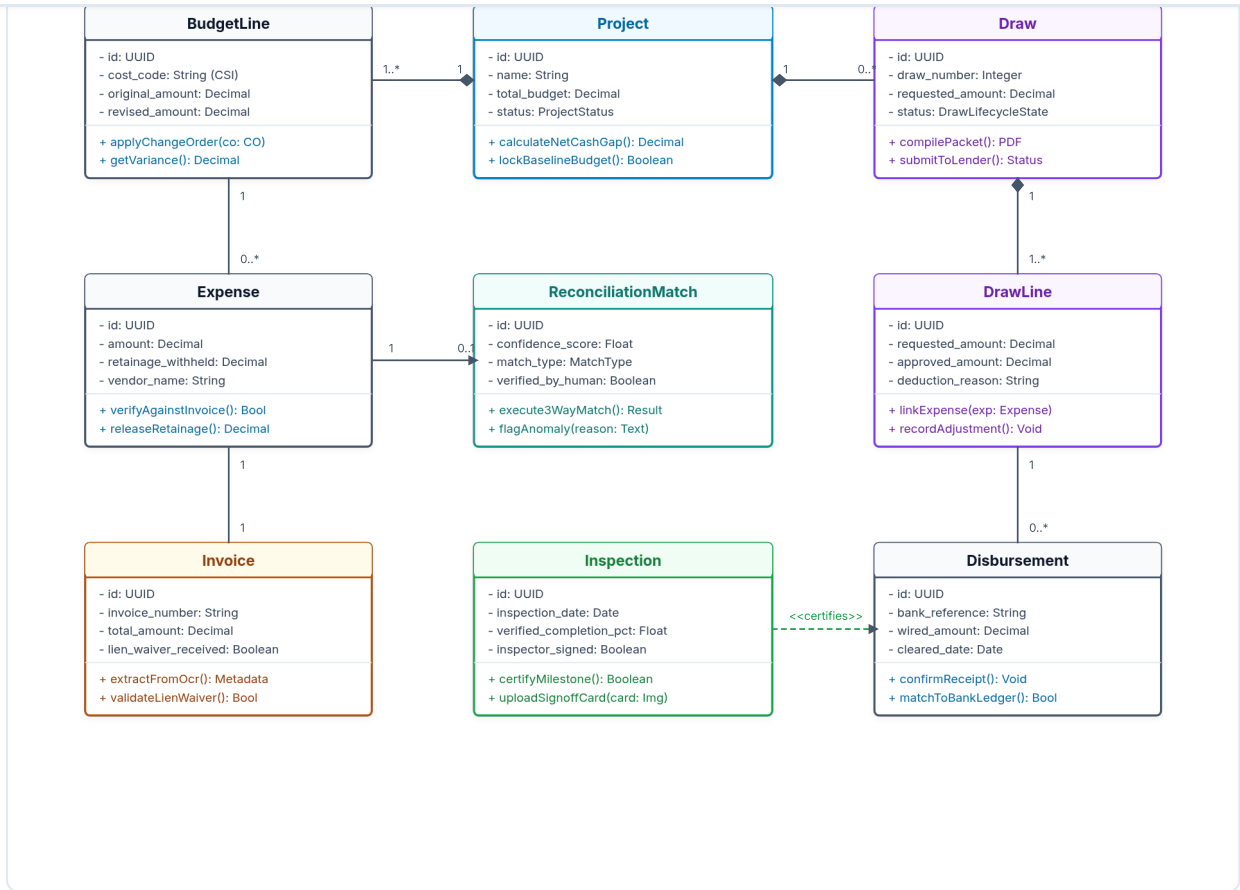


Figure 10: Class structure showing core fields, actions, and connections across project entities.

DATA FLOW TIMELINE

14. Invoice Ingestion to Draw Assembly Workflow

Step-by-step timeline showing what happens when a document is uploaded, scanned by AI, checked for accuracy, and saved to the budget.

FIGURE 11 Invoice Ingestion to Draw Assembly Workflow

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Live ERD Explorer (28 Tables)

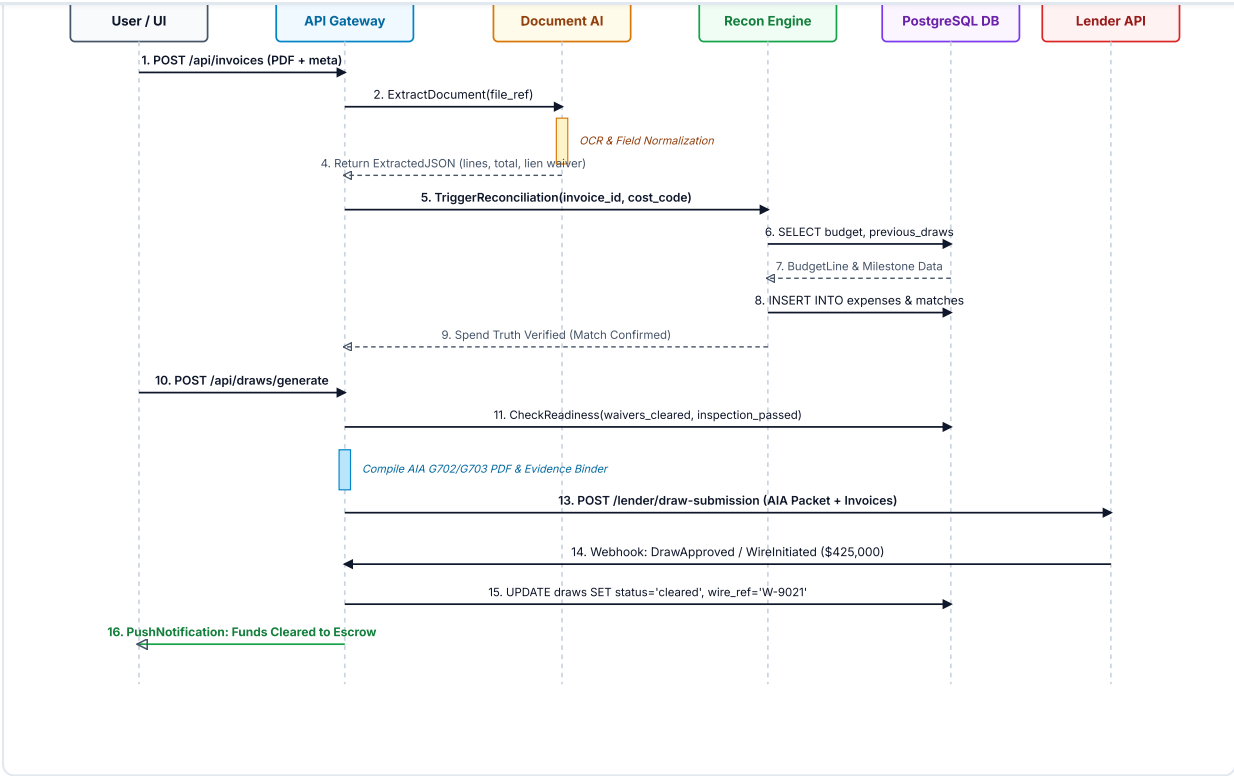


Figure 11: Interaction timeline showing document processing, AI extraction, and database persistence.

DATABASE STRUCTURE

15. Relational Database Architecture (ERD)

Explore Interactive ERD & Schema

Database structure showing how 15 core business tables connect together to provide an unbroken audit trail from land purchase to final investor payout. For live foreign key inspection and 28-table schema exploration, access the [Interactive ERD Explorer](#).

Live Interactive Database & ERD Explorer

Access the complete production database visualizer with 28 normalized tables, domain guides, and interactive foreign key relationships.

Open groundupdoc.vercel.app

 Live Embedded ERD Explorer (28 Tables) [LIVE APP](#)

[Open in New Tab](#)

GroundUp AI Specification

Live ERD Explorer (28 Tables)

Schema (33)

ERD Graph

PRD / TRD Spec

Download PDF

DATABASE TABLES

33

33 normalized schema models across system

FUNCTIONAL DOMAINS

10

Domain clusters and bounded operational contexts

RELATIONSHIPS

104

Foreign key relations connecting models & audit trails

DATA PRINCIPLES

8

Invariants ensuring audit trails and financial integrity

Search tables, fields (e.g. loan_id, project_id, jsonb, expenses)...

Group by Domain

All Grid

FILTER:

All Domains (33)

Organization & Team (3)

Projects & Economics (3)

Financing & C

Organization & Team

3 Tables

#01 organizations

ORGANIZATION & TEAM

id PK

uuid

Embedded Source: <https://groundupdoc.vercel.app/>
Click any entity or table node above to inspect foreign key links, schema definitions, and domain guides.

Static Relational Architecture (Offline Vector SVG Blueprint)

Permanent offline vector diagram illustrating the primary 15 business entities and relational foreign key connections.

FIGURE 12 Relational Database Architecture (ERD)

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<https://prdtrd.vercel.app>

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Live ERD Explorer (28 Tables)

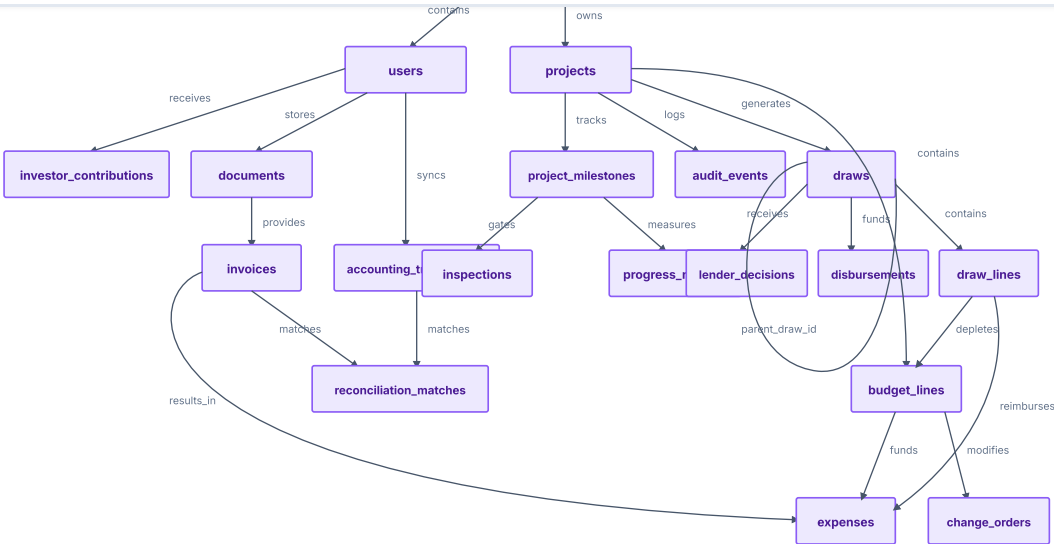


Figure 12: 15 interconnected database entities supporting Four Truths financial auditing and lifecycle tracking. For live schema exploration, use the embedded explorer above or visit <https://groundupdoc.vercel.app/>.

CORE DATA TABLES

Live ERD Explorer (28 Tables) →

16. Core Database Table Specifications

Master directory defining primary database records, key fields, and how each table supports real-world construction accounting.

TABLE	PRIMARY KEY	KEY FOREIGN KEYS	CRITICAL ATTRIBUTES
organizations	id (UUID)	None	name , corporate_domain , subscription_tier
users	id (UUID)	org_id	email , role , first_name , last_name , is_active

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projects	id (UUID)	org_id , created_by	name , code , site_address , lifecycle_stage , status
project_financial_baselines	id (UUID)	project_id	land_cost , hard_budget soft_budget , target_profit , target_irr
investor_contributions	id (UUID)	project_id , user_id	investor_name , equity_amount , contribution_date
budget_lines	id (UUID)	project_id	cost_code , category , baseline_budget , current_budget
change_orders	id (UUID)	project_id , budget_line_id , authorized_by	co_number , adjustment_amount , approved_date , status
contracts	id (UUID)	project_id , vendor_id	contract_number , contract_amount , retainage_rate , status

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Live ERD Explorer (28 Tables)

invoices	id (UUID)	project_id , contract_id	invoice_number , gross_amount , retainage_amount , net_payable_amount , status
payment_transactions	id (UUID)	project_id , invoice_id	account_source , transaction_type , amount_paid , transaction_date
draws	id (UUID)	project_id , parent_draw_id	draw_number , requested_amount , approved_amount , is_revision , status
draw_items	id (UUID)	draw_id , budget_line_id , invoice_id	requested_amount , approved_amount , line_status
milestones_timeline	id (UUID)	project_id , budget_line_id	milestone_name , baseline_end_date , forecast_end_date , is_completed
field_evidence	id (UUID)	milestone_id , draw_id	file_url , file_hash , inspection_type , signoff_date

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compliance_waivers	id (UUID)	invoice_id , vendor_id	waiver_type , through_date , waiver_amount , is_verified
unit_sales_disposition	id (UUID)	project_id	unit_number , sale_price broker_commission , net_proceeds , closing_date
project_realized_pl	id (UUID)	project_id	total_revenue , total_actual_costs , net_realized_profit , realized_irr , realized_em
investor_distributions	id (UUID)	project_id , user_id	distribution_amount , preferred_return_portion promote_portion , distribution_date
audit_trail_events	id (UUID)	user_id , project_id	entity_type , entity_id action_type , before_state , after_state

FINANCIAL ENGINEERING

17. Mathematical Logic: Live Pro Forma vs Current Forecast

The Fundamental Problem GroundUp Solves

Real estate developers fail when they rely on static spreadsheets that only compare paid invoices against original line-item budgets. In actual construction, projects suffer from two compounding financial vectors:

1. **Direct Scope Escalation:** Unforeseen soil conditions, design changes, and trade contractor change orders that directly deplete the contingency reserve.
2. **Schedule Delay and Daily Carrying Costs:** Construction loans (such as the \$4,200,000 Columbia Bank facility on 73 Broadway) accrue interest every single day. A 60-day delay does not just push completion back on a calendar; it generates 60 additional days of mandatory interest payments plus ongoing site general conditions (superintendent payroll, job trailer rental, temp power, site security).

METRIC / COMPONENT	MATHEMATICAL FORMULA	BASELINE VALUE (DAY 0 PLAN)	IMPACT OF DELAY AND OVERRUN
1. Baseline Total Cost (TDC)	$TDC_0 = Land + Hard + Soft + Contingency + Carry$	\$6,130,000 (73 Broadway Model)	Fixed underwriting anchor established on Day 0.
2. Forecast at Completion (FAC)	$FAC = Incurred_Spend + Remaining_Scope + Overrun + Added_Interest + Added_GC$	\$6,130,000	Expands dynamically as direct trade overruns and schedule delays accumulate.
3. Added Loan Interest	$Added_Interest = Drawn_Principal * (Rate / 365) * Delay_Days$	\$0 (at 0 days delay)	At \$4.2M loan and 8.25% interest rate, burns \$949.32 per day in added bank debt.
4. Added General Conditions	$Added_GC = (Monthly_GC_Rate / 30) * Delay_Days$	\$0 (at 0 days delay)	At \$12,500/month site overhead, burns \$416.67 per day in superintendent and trailer burn.
5. Forecast Net Profit	$Profit_Forecast = Projected_Revenue - FAC - Closing_Brokerage$	\$1,450,000	Decreases dollar-for-dollar with every dollar of cost overrun and daily debt carry.
6. Dynamic Investor IRR	$IRR = [(Invested_Equity + Profit_Forecast) /$	21.4% Annualized	Erodes severely due to the double penalty of

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Live ERD Explorer (28 Tables)

$$\frac{\text{Invested_Equity}}{(T_0 + \text{DeLay}/365)} - 1$$

reduced profit distributed over an extended timeline.

Real-World Case Study: 73 Broadway Overrun Simulation

If 73 Broadway encounters a **\$45,000 framing cost overrun** and a **40-day municipal inspection delay**:

- Added Columbia Bank interest: 40 days * \$949.32 = **+\$37,973**
- Added site general conditions: 40 days * \$416.67 = **+\$16,667**
- Total project cost escalation: \$45,000 + \$37,973 + \$16,667 = **+\$99,640**
- Current forecast profit drops from \$1,450,000 to **\$1,350,360**
- Investor IRR drops from **21.4%** to **18.1%**

Real-World Benchmark: 392 1st Street Disposition

The realized dataset from **392 1st Street** validates this exact model:

- Final unit sales proceeds matched against actual incurred costs from BCB bank feeds
- AMEX corporate card expenses reconciled directly against trade division lines
- Total realized developer profit confirmed upon final HUD closing distribution
- Actual LP equity multiple and realized IRR audited with zero spreadsheet gaps

API DESIGN

18. API Design and Integration Specifications

RESTful endpoints and webhook architectures connecting frontends, Document AI workers, and external banking feeds.

METHOD	ENDPOINT URI	PURPOSE & PAYLOAD	AUTHORIZED ROLES
POST	/api/v1/documents/upload	Uploads PDF invoices or bank statements; triggers asynchronous OCR processing.	GC, Super, Accountant
GET	/api/v1/projects/{id}/forecast	Returns baseline pro forma, actual incurred spend, daily interest carry, and forecast at completion.	Developer, Investor, Accountant

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POST	/api/v1/draws/{id}/assemble	Assembles draw packet, checks statutory California Civil Code Section 8132 waivers, and generates AIA G702/G703 PDF.	Accountant, Developer
POST	/api/v1/reconcile/bank-feed	Ingests BCB Bank wire transactions and AMEX credit card charges; matches to verified invoices.	Accountant, Super Admin
POST	/api/v1/disposition/unit-sales	Ingests executed unit sales agreements and gross closing proceeds (392 1st Street model).	Developer, Super Admin

RELIABILITY ENGINEERING

19. Technical Failure Recovery Matrix (9 Real-World Triggers)

Predefined system behaviors and deterministic recovery protocols for edge cases across extraction, banking, and field operations.

FAILURE TRIGGER	AFFECTED ENTITY	RESPONSIBLE ROLE	DETERMINISTIC SYSTEM RECOVERY PROTOCOL	RESULTING STATE
1. Low AI Confidence (< 0.85)	invoices	Accountant / Finance	Highlights low-confidence field in red in split-screen verification UI; blocks auto-post; prompts human confirmation.	CONFIRMED
2. Unmatched Ingestion File	documents	Accountant / Finance	Routes document to Project Ingestion Queue; isolates	VERIFIED

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Live ERD Explorer (28 Tables)

PROTOCOL				
from Spend Truth until manual cost-code assignment.				
3. Suspected Duplicate Invoice	invoices	Accountant / Finance	Detects identical vendor and invoice number; presents side-by-side comparison modal; prevents double billing.	FLAGGED_DUPL
4. Missing Statutory Lien Waiver	compliance_waivers	General Contractor	Verifies absence of statutory California Civil Code Section 8132 waiver; displays warning; withholds line from draw packet.	PENDING_WAIV
5. Failed City Inspection	field_evidence	Project Superintendent	Municipal inspector flags deficiency on framing card; freezes only the affected cost code; unaffected trades proceed.	RE_INSPECTIO
6. Lender Draw Deduction (\$10K held)	draw_items	Developer / Owner	Columbia Bank cuts \$10K from draw; system accepts approved funds and auto-spawns a child revision draw	DEFERRED_TO_

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Live ERD Explorer (28 Tables)

PROTOCOL				
for the withheld item.				
7. Full Draw Packet Rejection	draws	Developer / Owner	Preserves immutable audit trail of bank rejection; clones items into a new revision packet with remediation notes.	DRAFT_REVISION
8. Bank Wire Discrepancy	payment_transactions	Accountant / Finance	Disbursed wire amount differs from approval letter; logs deposited cash; creates auto-balancing adjustment entry.	DISCREPANCY_TRANSACTION
9. Budget Line Cost Overrun	budget_lines	Developer / Owner	Invoice line exceeds remaining current budget; blocks automated post; prompts formal change order or contingency transfer.	CHANGE_ORDER_REQUEST

REFERENCE REGISTER

20. Citations, Statutory References, and Terminology Appendix

Official statutory citations, industry standards, financial terms, and multi-project definitions governing the GroundUp AI platform.

Part A: Statutory & Industry Standards Citations

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Live ERD Explorer (28 Tables)

California Civil Code Section 8132	State of California	Mandatory conditional lien waiver form upon progress payment; enforced prior to invoice draw inclusion.
California Civil Code Section 8134	State of California	Mandatory unconditional lien waiver form upon progress payment; releases lien rights once payment clears.
California Civil Code Section 8136 & 8138	State of California	Statutory final conditional and unconditional waiver forms required prior to final 10% retainage release.
AIA Document G702 / G703	American Institute of Architects	Standard Application and Certificate for Payment with Continuation Sheet; standard format for bank draw submission.
AIA Document A201-2017	American Institute of Architects	General Conditions of the Contract for Construction; establishes standard 10% retainage holdback rules.
CSI MasterFormat (50 Divisions)	Construction Specifications Institute	Standardized work breakdown structure mapping budget lines, contracts, invoices, and draws across trades.

Part B: Financial & Real Estate Terminology Appendix

TERM	FORMAL INDUSTRY DEFINITION	GROUNDUP OPERATIONAL ROLE
The Four Truths	The four synchronized financial records of a project: Budget Truth, Spend Truth, Progress Truth, and Funding Truth.	Core data engine connecting pro forma baselines, approved vendor expenses, physical site completion, and bank loan draws.
Forecast at Completion (FAC)	Estimated total cost of the project upon completion, combining actuals, remaining scope, and delay debt carry.	Dynamic KPI calculated live: Incurred Costs + Remaining Balance + Overrun + Added Carry.
Retainage (Retention)	A portion of agreed contract price (typically 10%) withheld until work is substantially completed.	Automated calculation on gross invoices; isolated into retainage payable ledger until final release.

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Daily Debt Carry Burn	Accrued daily interest on drawn loan principal resulting from construction schedule delay.	Calculated as: Drawn Balance multiplied by (APR / 365) multiplied by Delay Days (\$949.32/day on \$4.2M).
Internal Rate of Return (IRR)	Annualized rate of earnings generated by an investment over its duration; heavily sensitive to time.	Dynamic KPI modeling how construction delays discount returns over an extended schedule.
Equity Multiple (EM)	Total cash distributions returned divided by total equity invested.	Tracks cash return ratio: (Invested Capital + Net Profit) / Invested Capital.

Part C: Multi-Project Asset Directory

- 73 Broadway:** Active ground-up construction asset; primary validation model for Columbia Bank construction loan draw statements, BCB Bank operational transfers, and monthly draw packets.
- 392 1st Street:** Completed residential disposition asset; validation model for condo unit sales contracts, selling prices, P&L cost per unit, AMEX corporate card statements, and realized investor distributions.
- 161 Woodlawn:** Pipeline pre-development asset; validation model for initial budget sheets, pre-construction expenses, and multi-tenant project onboarding.